

4.4 SUMMARY

In this chapter, we have derived equations for the propagation of the mean and covariance of the state of linear systems. For discrete-time systems, the mean and covariance are described by difference equations. Sampled-data systems are systems with continuous-time dynamics but control inputs that are constant between sample times. If the dynamics of a sampled-data system does not change between sample times, then the mean and covariance are described by difference equations, although the factors of the difference equations are more complicated than they are for discrete-time systems. For continuous-time systems, the mean and covariance are described by differential equations. These results will form part of the foundation for our Kalman filter derivation in Chapter 5.

The covariance equations that we studied in this chapter are named after Aleksandr Lyapunov, James Sylvester, and Philip Stein. Lyapunov was a Russian mathematician who lived from 1857 to 1918. He made important contributions in the areas of differential equations, system stability, and probability. Sylvester was an English mathematician and lawyer who lived from 1814 to 1897. He worked for a time in the United States as a professor at the University of Virginia and Johns Hopkins University. While at Johns Hopkins, he founded the *American Journal of Mathematics*, which was the first mathematical journal in the United States.

PROBLEMS

Written exercises

4.1 Prove that

$$\frac{d}{dt}(E[x]) = E\left[\frac{dx}{dt}\right]$$

4.2 Suppose that a dynamic scalar system is given as $x_{k+1} = fx_k + w_k$, where w_k is zero-mean white noise with variance q . Show that if the variance of x_k is σ^2 for all k , then it must be true that $f^2 = (\sigma^2 - q)/\sigma^2$.

4.3 Consider the system

$$\begin{aligned} x_k &= \begin{bmatrix} 1 & 1 \\ 0 & 1/2 \end{bmatrix} x_{k-1} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} w_{k-1} \\ w_k &\sim (0, 1) \end{aligned}$$

where w_k is white noise.

- a) Find all possible steady-state values of the mean of x_k .
- b) Find all possible steady-state values of the covariance of x_k .

4.4 Consider the system of Example 1.2.

- a) Discretize the system to find the single step state transition matrix F_k , the discrete-time input matrix G_k , and the multiple-step state transition matrix $F_{k,i}$.

- b) Suppose the covariance of the initial state is $P_0 = \text{diag}(1, 0)$, and zero-mean discrete-time white noise with a covariance of $Q = \text{diag}(1, 0)$ is input to the discrete-time system. Find a closed-form solution for P_k .

4.5 Two chemical mixtures are poured into a tank. One has concentration c_1 and is poured at rate F_1 , and the other has concentration c_2 and is poured at rate F_2 . The tank has volume V , and its outflow is at concentration c and rate F . This is typical of many process control systems [Kwa72]. The linearized equation for this system can be written as

$$\dot{x} = \begin{bmatrix} -\frac{F_0}{2V_0} & 0 \\ 0 & -\frac{F_0}{V_0} \end{bmatrix} x + \begin{bmatrix} 1 & 1 \\ \frac{c_1 - c_0}{V_0} & \frac{c_2 - c_0}{V_0} \end{bmatrix} w$$

where F_0 , V_0 , and c_0 are the linearization points of F , V , and c . The state x consists of deviations from the steady-state values of V and c , and the noise input w consists of the deviations from the steady-state values of F_1 and F_2 . Suppose that $F_0 = 2V_0$, $c_1 - c_0 = V_0$, and $c_2 - c_0 = 2V_0$. Suppose the noise input w has an identity covariance matrix.

- Use Equation (4.27) to calculate Q_{k-1} .
- Use Equation (4.28) to approximate Q_{k-1} .
- Evaluate your answer to part (a) for small $(t_k - t_{k-1})$ to verify that it matches your answer to part (b).

4.6 Suppose that a certain sampled data system has the following state-transition matrix and approximate Q_{k-1} matrix [as calculated by Equation (4.28)]:

$$\begin{aligned} F_{k-1} &= \begin{bmatrix} e^{-T} & 0 \\ 0 & e^{-2T} \end{bmatrix} \\ Q_{k-1} &= \begin{bmatrix} 2T & 3T \\ 3T & 5T \end{bmatrix} \end{aligned}$$

where $T = t_k - t_{k-1}$ is the discretization step size. Use Equation (4.26) to compute the steady-state covariance of the state as a function of T .

4.7 Consider the tank system described in Problem 4.5. Find closed-form solutions for the elements of the state covariance as functions of time.

4.8 Consider the system

$$\begin{aligned} x_{k+1} &= \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} x_k + w_k \\ w_k &\sim (0, Q) \\ Q &= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

Use Equation (4.5) to find the steady-state covariance of the state vector.

4.9 The third condition of Theorem 21 gives a sufficient condition for the discrete-time Lyapunov equation to have a unique, symmetric, positive semidefinite solution. Since the condition is sufficient but not necessary, there may be cases that do not meet the criteria of the third condition that still have a unique, symmetric, positive semidefinite solution. Give an example of one such case with a nonzero solution.

4.10 Prove the sifting property of the continuous-time impulse function $\delta(t)$, which can be stated as

$$\int_{-\infty}^{\infty} f(t)\delta(t - \alpha) dt = f(\alpha)$$

Computer exercises

4.11 Write code for the propagation of the mean and variance of the state of Example 4.2. Use $m_0 = 1$, $P_0 = 2$, $f = -0.5$ and $q_c = 1$. Plot the mean and variance of x for 5 seconds. Repeat for $P_0 = 0$. Based on the plots, what does the steady-state value of the variance appear to be? What is the analytically determined steady-state value of the variance?

4.12 Consider the RLC circuit of Example 1.8 with $R = L = C = 1$. Suppose the applied voltage is continuous-time zero-mean white noise with a variance of 1. The initial capacitor voltage is a random variable with a mean of 1 and a variance of 1. The initial inductor current is a random variable (independent of the initial capacitor voltage) with a mean of 2 and a variance of 2. Write a program to propagate the mean and covariance of the state for five seconds. Plot the two elements of the mean of the state, and the three unique elements of the covariance. Based on the plots, what does the steady-state value of the covariance appear to be? What is the analytically determined steady-state value of the covariance? (Hint: The MATLAB function LYAP can be used to solve for the continuous-time algebraic Lyapunov equation.)

4.13 Consider the RLC circuit of Problem 1.18 with $R = 3$, $L = 1$, and $C = 0.5$. Suppose the applied voltage is continuous-time zero-mean white noise with a variance of 1. We can find the steady-state covariance of the state a couple of different ways.

- Use Equation (4.49).
 - Discretize the system and use Equation (4.4) along with the MATLAB function DLYAP. In this case, the discrete-time white noise covariance Q is related to the continuous-time white noise covariance Q_c by the equation $Q = TQ_c$, where T is the discretization step size (see Section 8.1.1).
- a) Analytically compute the continuous-time, steady-state covariance of the state.
 - b) Analytically compute the discretized steady-state covariance of the state in the limit as $T \rightarrow \infty$.
 - c) One way of measuring the distance between two matrices is by using the MATLAB function NORM to take the Frobenius norm of the difference between the matrices. Generate a plot showing the Frobenius norm of the difference between the continuous-time, steady-state covariance of the state, and the discretized steady-state covariance of the state for T between 0.01 and 1.

PART II

THE KALMAN FILTER
